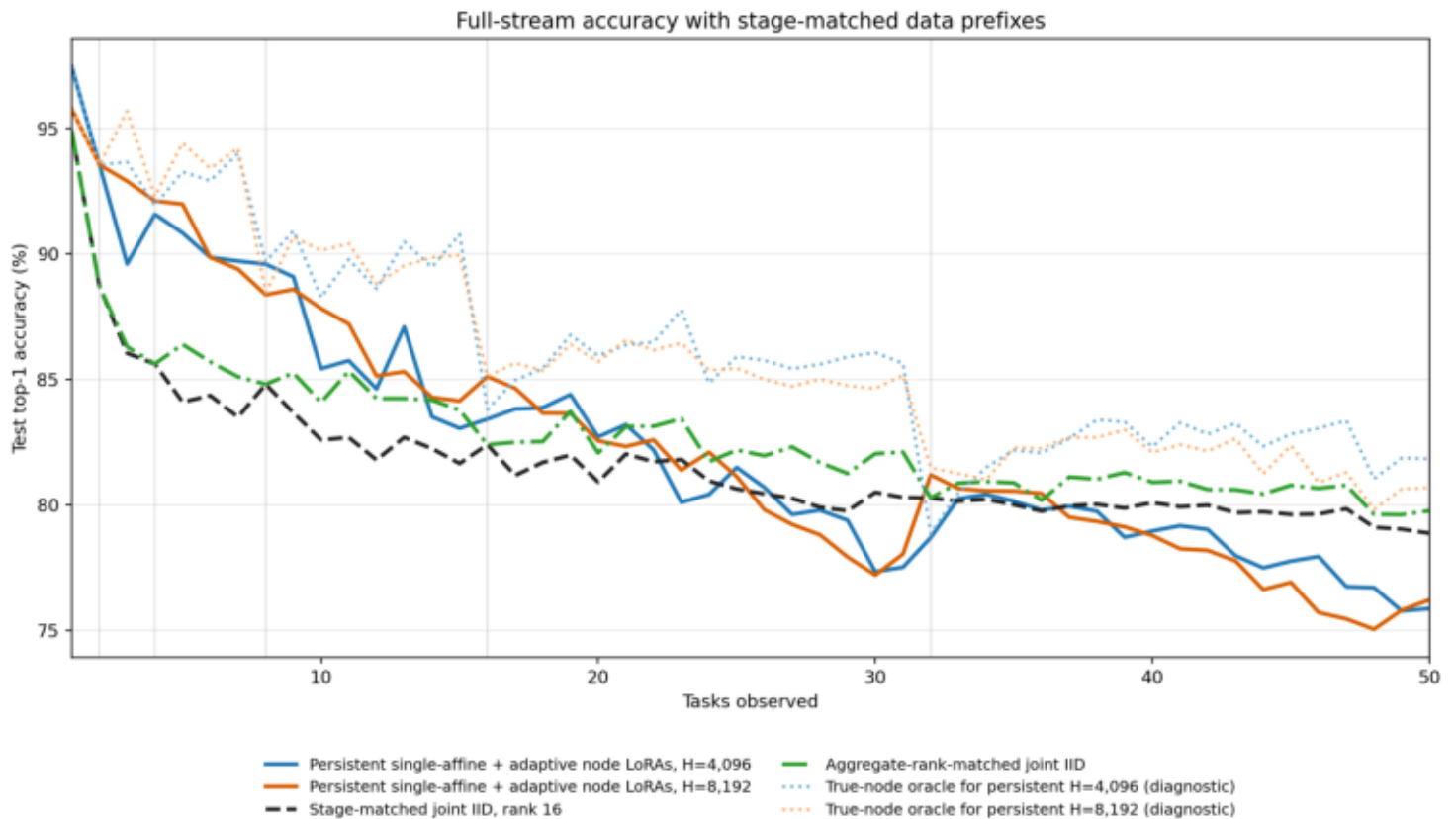


ImageNet-R-50 persistent single-affine frontier

Full 50-task extension of the H=4,096 and H=8,192 adaptive-LoRA conditions

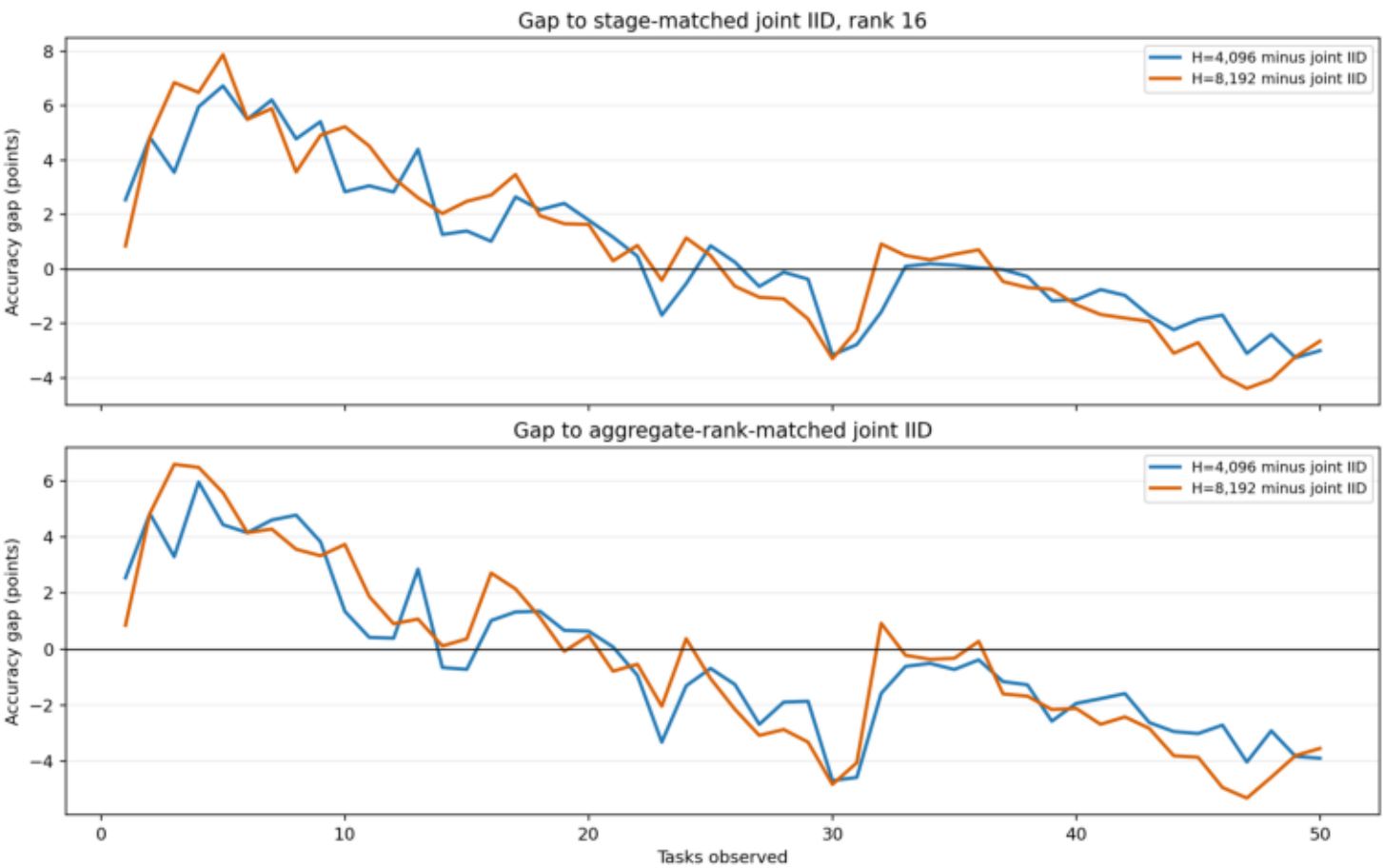
At task 50, H=4,096 reaches 75.867% and H=8,192 reaches 76.217%. The corresponding rank-16 and aggregate-rank joint-IID references are 78.867% and 79.767%. Across all 50 stages, incremental accuracy is 82.434% for H=4,096 and 82.452% for H=8,192, versus 81.630% and 82.759% for the two joint-IID curves.



The adaptive curves use only one affine layer over node-specific pre-classifier vectors. Their dotted true-node curves are label-aware diagnostics, not deployable conditions. The black control is fresh rank-16 joint IID at each prefix. The green control uses rank 16 times the number of live nodes.

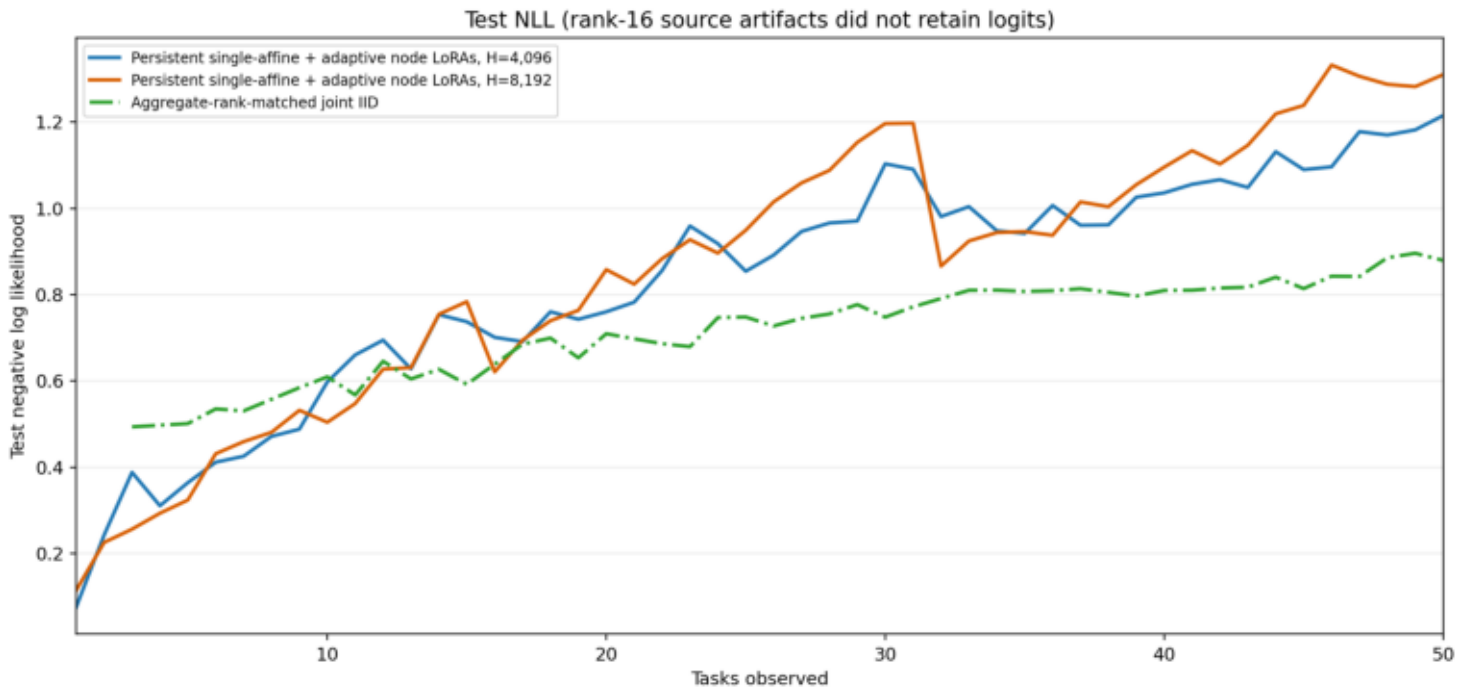
Selected stages and control gaps

Stage	Nodes	H=4,096	H=8,192	H4 oracle	H8 oracle	Joint r16	Rank-matched	Rank
1	1	97.46%	95.76%	97.46%	95.76%	94.92%	94.92%	16
2	1	93.55%	93.55%	93.55%	93.55%	88.71%	88.71%	16
4	1	91.58%	92.11%	91.93%	92.28%	85.61%	85.61%	16
8	1	89.59%	88.37%	89.68%	88.46%	84.80%	84.80%	16
16	1	83.41%	85.11%	83.80%	85.11%	82.40%	82.40%	16
31	5	77.52%	78.04%	85.64%	85.17%	80.29%	82.10%	80
32	1	78.69%	81.19%	78.85%	81.47%	80.28%	80.28%	16
50	3	75.87%	76.22%	81.83%	80.68%	78.87%	79.77%	48



Positive gap means the persistent frontier is ahead of the named joint-IID reference. Power-of-two stages have one node; fragmented stages have two or more.

Likelihood and protocol



Rank-16 source artifacts retained hard predictions rather than logits, so rank-matched NLL is absent at stages 1, 2, 4, 8, 16, and 32. Accuracy remains available at all 50 stages.

Exact online update rule

Every arrival uses all current-task images plus a deterministic stage-keyed class-stratified historical draw. $H=4,096$ receives four passes and $H=8,192$ receives five. AdamW state and affine parameters carry forward. A node adapter and its moments carry only if its hierarchy-node hash survives; new leaves and consolidation parents start from authenticated source adapters. The base ViT and local classifiers never train.

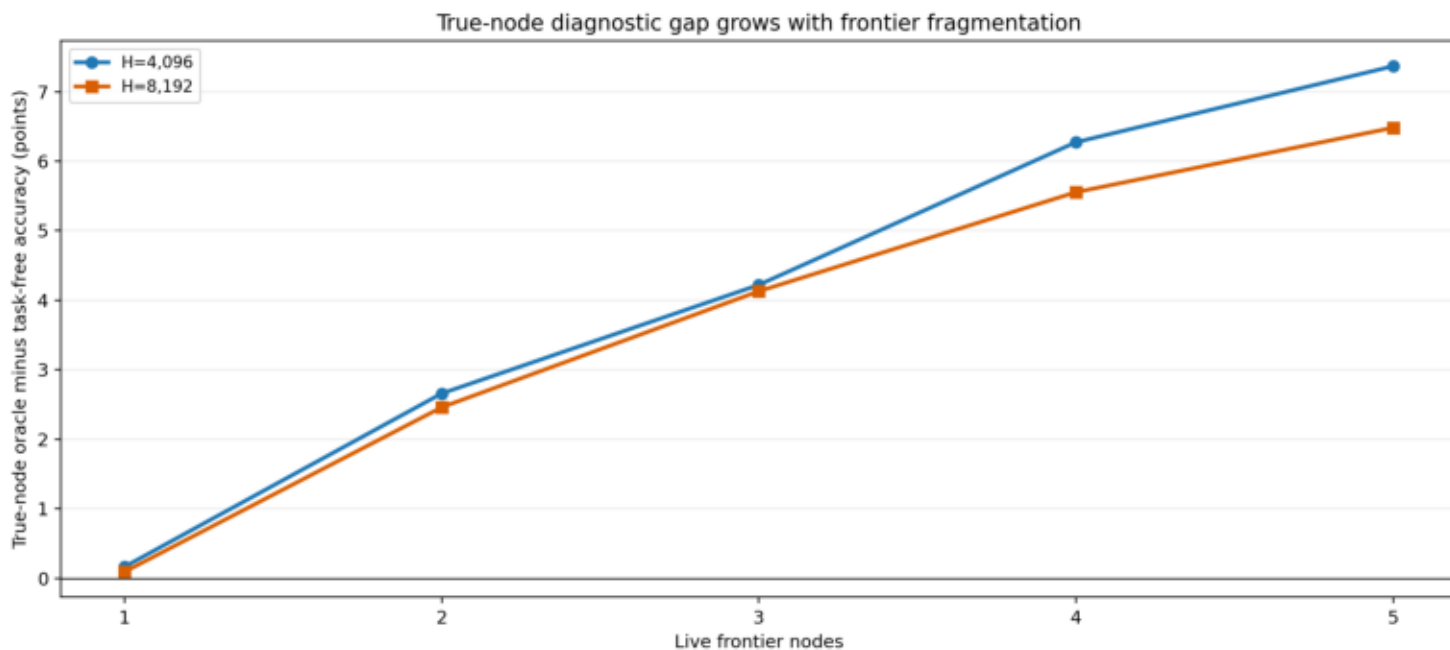
Comparison boundary

Stage-matched joint IID trains one fresh rank-16 adapter for five epochs on every prefix. Aggregate-rank-matched joint IID uses rank and alpha equal to 16 times popcount(stage). It matches total live LoRA rank, but not pretrained node state, multiple ViT paths, affine-head parameters, or deployment compute. Neither curve gates execution.

Provenance

Protocol 5f60e2f5c25e3318ba9c9b66d41ce8c159ba57a0f1a4fc524567c098b222d686
Result 4fa601c9b8e3cc87f07897f06ba51f88b75b7ecde1bf47ef01881526df4b4b58
Source hierarchy unchanged: True

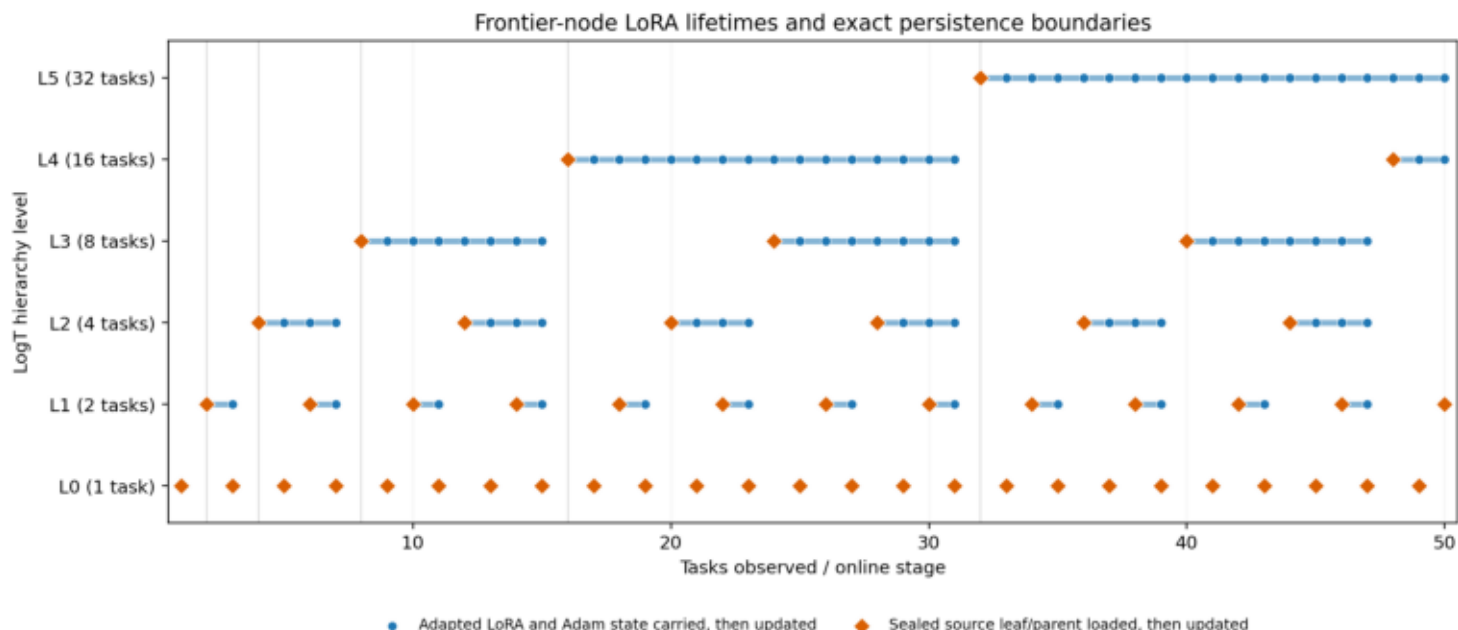
Fragmentation diagnosis



Nodes	Stages	H4 acc.	H4 gap	H8 acc.	H8 gap
1	6	89.05%	0.16	89.35%	0.09
2	15	83.76%	2.66	84.31%	2.46
3	18	81.39%	4.22	81.05%	4.13
4	9	79.10%	6.27	78.82%	5.55
5	2	77.13%	7.37	76.75%	6.48

The label-aware true-node oracle is diagnostic only. Its growing advantage as the number of independently adapted frontier nodes rises is consistent with cross-node competition. Because it also substitutes the frozen node-local classifier, it does not isolate routing alone.

Frontier-node adaptation lifetimes



What persists

A blue lifetime continues only while the exact source hierarchy-node hash remains live. The next stage begins from that node's final adapted rank-16 LoRA, its 768-column affine block, and its named AdamW moments. All live LoRAs then update jointly on every current-plus-replay example.

What resets

An orange diamond is a new leaf or consolidation parent loaded from the sealed source hierarchy. A parent was independently retrained on the full union of its represented tasks; it does not inherit the online-adapted LoRAs or optimizer moments of retired children. Unaffected nodes continue normally, and old global affine-bias rows remain persistent.